

Checklist for assessment of GSP pipelines

Version	Date
v1	2026-02-02

Purpose

This document provides a checklist of important features that are required for Nextflow pipelines prior to integration into the Canadian National Microbiology Laboratory's Genomic Surveillance Platform (GSP). This document is intended to be used by pipeline developers to help evaluate their pipelines for readiness to integrate into GSP or to provide guidance on requirements for new pipelines. The checklist follows the structure of the NMLs Pipeline Standards document (<https://github.com/phac-nml/pipeline-standards>), which can be referred to for additional information.

Usage

First, review your pipeline and check off "Yes" or "No" in each box for each of the questions. Any unmet criteria marked as *Required* will need to be resolved prior to the pipeline being integrated into GSP. Recommended actions are not required but improve integration into GSP.

For clarification, contact the Bioinformatics Pipelines Development unit or Aaron Petkau <aaron.petkau@phac-aspc.gc.ca>.

Pipeline evaluation

Pipeline name: Click or tap here to enter text.

Pipeline author: Click or tap here to enter text.

Pipeline code repository: Click or tap here to enter text.

Question	Required or Recommended	If answer is 'no', how to resolve
1. Pipeline code repository		
☐ 1.1. Does the pipeline provide a "main" branch and a "dev" development branch in the git repository?	Recommended	It's recommended to have a "dev" branch for in-development code and a "main" branch for maintaining pipeline releases.

□	1.2. Was the pipeline created using our iridanextexample (or iridanextexample2) git repository as a template or by using the “nf-core pipelines create” command from the nf-core tools?	Recommended	We recommend starting from the iridanextexample (or iridanextexample2) git repository as a template for your pipeline, as we have made some custom modifications to these repos to remove nf-core specific branding. Alternatively, you can use the “nf-core pipelines create” command from nf-core tools to create a pipeline template, but you then will have to make additional changes to conform to our GSP pipeline requirements. See our Pipeline Standards documentation for which requirements we use and don’t use from nf-core. <i>Note: If you are just starting out creating a brand new pipeline, we would recommend contacting the Bioinformatics Pipelines Development unit (or Aaron Petkau). We are in the process of updating IRIDA Next to support more recent nf-core pipeline standards, which are captured by the iridanextexample2 pipeline template. Until the IRIDA Next work is complete, pipelines based on iridanextexample2 won’t load in IRIDA Next, but we would still recommend starting development from that repo.</i> <i>Additionally, until updates to IRIDA Next are made, if you use the latest version of the “nf-core” commands to create a pipeline template, you will have to downgrade the Nextflow plugin “nf-schema” to “nf-validation”, and the version of the JSON schemas for “nextflow_schema.json” and “schema_input.json” files to match the “iridanextexample” pipeline due to incompatibilities with IRIDA Next. Please see (2.3 and 2.4) for more information.</i>
2. Input			
□	2.1. Does the pipeline accept a <i>samplesheet.csv</i> file with sample data using the “--input” parameter?	Required	Provide a parameter “--input” that can be used to pass a <i>samplesheet.csv</i> with input data. The structure of the input CSV file is defined by the <i>schema_input.json</i> JSON schema defined for each pipeline (see 2.4).

☐	2.2. Does the pipeline accept a <i>samplesheet.csv</i> with both a column named <i>sample</i> and a column named <i>sample_name</i> ?	Recommended	It's recommended to have an input column named <i>sample</i> and one named <i>sample_name</i>. The <i>sample</i> column is filled in by IRIDA Next with the PUID of a sample (e.g., <i>INXT_SAM_ABC123</i>), while the <i>sample_name</i> column is filled in with the assigned sample name (e.g., <i>26-0001</i>). The pipeline will continue to work if only a column with <i>sample</i> is available, but sample names will be unavailable for inclusion in outputs of a pipeline in this case.
☐	2.3. Does the pipeline use the <i>nf-validation</i> Nextflow plugin to validate the input <i>samplesheet.csv</i> and parameters and create channels of input data?	Required	The <i>nf-validation</i> Nextflow plugin (https://nextflow-io.github.io/nf-validation/) is required to validate the input <i>samplesheet</i>, parameters, and construct the channels used by Nextflow for downstream processing. Note: <i>nf-validation</i> has been replaced by the <i>nf-schema</i> Nextflow plugin, but IRIDA Next and GSP do not yet support <i>nf-schema</i>. Support is coming soon, so please reach out to the Pipeline Development unit for up-to-date information.
☐	2.4. Does the pipeline provide an "assets/schema_input.json" that reflects input in the <i>samplesheet.csv</i> ?	Required	Provide an "assets/schema_input.json" that reflects the input <i>samplesheet</i>. This JSON schema file is used by the Nextflow <i>nf-validation</i> plugin (and the updated <i>nf-schema</i> plugin), and by IRIDA Next to render the user interface for submitting a pipeline.
☐	2.5. Are all necessary database files/folders able to be passed into the pipeline via parameters (like "--database")?	Required	Provide parameters for all necessary databases/required data files that can be used to point to paths of data (e.g., <code>--kraken2_db /kraken2/db</code>). In Azure, we store necessary data for a pipeline in Azure blob

			storage and pass paths to this storage via parameters (e.g., <code>--kraken2_db az://kraken2/db</code>). In IRIDA Next, we configure the pipeline to represent these necessary databases/files as drop-down menus. See Pipeline Standards: Referencing External Data .
☐	2.6. If the pipeline has the ability to auto-download necessary data, is there a provided mechanism to disable auto-downloading?	Required	Provide a mechanism to disable auto-downloading data (e.g., a flag like <code>--disable-autodownload</code>) so that the pipeline will always run with the same versions of necessary data/databases and so that multiple runs of the pipeline won't continue to re-download the same data.
3. Parameters			
☐	3.1. Does the pipeline provide a <code>nextflow_schema.json</code> JSON Schema of parameters that is in-sync with " <code>nextflow.config</code> " parameters?	Required	Provide a " <code>nextflow_schema.json</code> " file that lists the parameters that are defined in the " <code>nextflow.config</code> " file. This JSON schema file is used by the Nextflow <i>nf-validation</i> plugin, and is used by IRIDA Next to render the pipeline submission user interface.
4. Outputs			
☐	4.1. Does the pipeline write output files using a directory configured with " <code>--outdir</code> "?	Required	All output files are required to be written to a single output directory configurable with " <code>--outdir</code> ".
☐	4.2. Does the pipeline make use of the " <i>nf-iridanext</i> " plugin to construct an IRIDA Next JSON file of outputs?	Required	The " <i>nf-iridanext</i> " plugin is required to construct IRIDA Next JSON files, which maintains consistency across all pipelines and helps with future updates to the file format.

5. Resource configuration			
☐	5.1. Does the pipeline define the following resource labels and designate each (non-“exec”) process with one of them: <i>process_single</i> , <i>process_low</i> , <i>process_medium</i> , <i>process_high</i> , <i>process_very_high</i> ?	Required	Required to provide resource labels for all processes in the pipeline as these are used to match to appropriate computational resources in GSP. Note: Processes with an “exec” code block will be executed by Nextflow directly and so don’t need a resource label.
☐	5.2. Do all processes (other than “exec” processes) have a docker container configured?	Required	Required to have docker containers for all processes as docker is used in GSP. It is recommend to use containers hosted via <i>quay.io/biocontainers</i>. Another option is Seqera Containers (https://seqera.io/containers/). Note: Processes with an “exec” code block will be executed by Nextflow directly and so don’t need a docker container configured.
☐	5.3. Do container strings contain any server prefixes (e.g., “quay.io” or “docker.io”)?	Required	Required to remove all server prefixes from container strings and instead set “<i>docker.registry</i>” with the main container server prefix. This is so we can configure a private container registry within GSP for caching containers. For pipelines using containers from multiple container registries, see the Pipeline Standards: Alternate Container Registries documentation.
☐	5.4. Does the pipeline provide the parameters “--max_cpus”, “--max_memory”, and “--max_time” for limiting resources?	Recommended	These parameters are used by Nextflow configuration to adjust the maximum resources used by pipeline processes in GSP.
6. Error handling			
☐	6.1. Are there any recoverable errors in the pipeline that will cause a failure?	Recommended	Consider adjusting to provide an error file as output in this situation. See the Pipeline Standards Error Handling documentation.

7. Testing			
☐	7.1. Does the pipeline contain a "test" profile for running with test data?	Required	Provide a "test" profile so that the pipeline can be tested on the command-line.
☐	7.2. Does the pipeline include automated tests (using "nf-test")?	Required	Required to add at least one full pipeline test case by making use of the nf-test software.
☐	7.3. Does the pipeline pass linting using the nf-core tool (nf-core pipelines lint)?	Required	We recommend making use of the nf-core linting tool to test the pipeline for conformance to nf-core standards. See the nf-core lint documentation for how to disable certain linting checks if they aren't needed by your pipeline.
8. Continuous integration			
☐	8.1. Are automated tests configured to be run as GitHub actions for pull requests?	Required	Setting up automated tests in the pipeline git repository allows for newer bug fixes and updates to be tested prior to pipeline release. It's recommended to make use of the GitHub actions files provided by the nf-core pipelines template and adjust as needed.
☐	8.2. Does the pipeline run "nf-core pipelines lint" as a GitHub action?	Required	A GitHub action for running "nf-core pipelines lint" automates evaluation of newer bug fixes and updates against nf-core standards. It's recommended to make use of the GitHub actions files provided by the nf-core pipelines template and adjust as needed.
9. Publishing			
☐	9.1. Does the pipeline contain a copyright notice "Copyright Government of Canada" or a similar statement?	Required	Add a copyright statement to the GitHub repository (in the README file) as this is required for code published by PHAC. See the Government of Canada Guide for Publishing Open Source Software for more information.

☐	9.2. Does the pipeline provide a <i>LICENSE</i> file and use an open source license?	Required	Select an open-source license (recommended MIT) and provide the LICENSE file in the git repository to document how others can use the code. See the Government of Canada Guide for Publishing Open Source Software for more information.
☐	9.3. Has all private data (if any) been removed from the git history prior to a public release?	Required	Remove any private data that should not be released, both in the latest code and in the git history. This could involve copying the latest code into a brand-new git repository, or modifying the git history.
☐	9.4. Does the pipeline provide a version number in “nextflow.config” and tag the released version in git?	Required	Provide a version in the “manifest” section of the “nextflow.config” and, when ready for release, tag the git <i>main</i> branch with this version. Versions of pipelines in GSP are linked to versions of code using the git tag and also the nextflow.config. We would recommend following semantic versioning for pipeline releases: https://semver.org/
10. Running the pipeline			
☐	10.1. Can the pipeline run with Nextflow version 24.10.X?	Required	Fix any issues that may be encountered when running with Nextflow 24.10.X. This is the current compatible version of Nextflow for GSP.
☐	10.2. Can the pipeline run using Azure Batch?	Required	Test out the pipeline in GSP using Azure Batch and fix any issues that come up. Please contact the GSP pipelines group for information on how to test the pipeline in Azure. Testing this out will be one of the last actions to take as most of the other items will need to be addressed first.

Resources

Additional resources for pipeline/software development can be found below.

- **NML Pipeline Standards for GSP:** <https://github.com/phac-nml/pipeline-standards>
- **Existing GSP Pipelines:** <https://github.com/phac-nml/nf-pipelines>
- **iridanextexample pipeline template:** <https://github.com/phac-nml/iridanextexample>
- **iridanextexample2 pipeline template:** <https://github.com/phac-nml/iridanextexample2>
- **nf-core:** <https://nf-co.re/>
- **nf-core tools:** <https://nf-co.re/docs/nf-core-tools>
- **nf-iridanext:** <https://github.com/phac-nml/nf-iridanext>
- **nf-test:** <https://www.nf-test.com/>
- **nf-validation:** <https://nextflow-io.github.io/nf-validation/>
- **nf-schema:** <https://nextflow-io.github.io/nf-schema/>
- **Seqera Containers:** <https://seqera.io/containers/>
- **Government of Canada Guide for Publishing Open Source Software:** [Guide for Publishing Open Source Code - Canada.ca](#)
- **Government of Canada Open by Default:** [Work in the open by default - Canada.ca](#)